

Trigonometric Functions

9

SYLLABUS 3.5–3.11

Your voice is a wave. So is the light from your screen, the signal carrying this page to your device, the current in the wall socket, the tide in Tokyo Bay, and the beating of your own heart.

They do not look alike. A heartbeat is a sharp spike, a spoken vowel is jagged, a tide is a slow rise and fall over twelve hours. In 1822 Joseph Fourier made a startling claim: a repeating signal, however jagged, can be built up from pure sine waves. Choose enough of them, with the right heights and the right timings, and the shape appears. Your phone uses this idea many times a second to compress your voice.

So the same curve describes all of them. Where does it come from?

Watch one seat on a turning Ferris wheel. As the wheel turns steadily, the seat rises, slows near the top, falls, slows near the bottom, and rises again. Plot its height against time and the sine curve appears. The seat is a point moving round a circle, and its height is the vertical coordinate of that point. The circle and the wave are the same motion seen two ways: one going round, one going along.

That is why this chapter starts with a circle. In Ch. 8 sine and cosine were ratios inside a right-angled triangle, so the angle had to be acute. On the circle they become functions of any angle, and they repeat forever.

By the end of this chapter you will be able to read exact values of sine, cosine and tangent from the unit circle. You will use identities to rewrite one expression as another, sketch and transform the graphs of these functions, and work with the reciprocal and inverse functions. You will also solve trigonometric equations, where the answers come in families: infinitely many solutions described by a single formula.

9.1 The Unit Circle

You already have a definition. In a right triangle,

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

This works, but only for acute angles. The three angles of a triangle add to 180° , and one of them is already 90° , so θ must be less than 90° . There is no right triangle containing an angle of 120° , or 400° , or -50° . Under this definition $\sin 120^\circ$ has no meaning at all.

Yet a rotation of 120° is perfectly ordinary. We need a wider definition, one that still agrees with the old one wherever the old one works.

The unit circle gives it. Take the circle of radius 1 centred at the origin, and a point P on it at angle θ , measured anticlockwise from the positive x -axis.

For an acute θ , drop a perpendicular from P to the x -axis. This makes a right triangle whose hypotenuse is the radius, so the hypotenuse is 1. The old ratios then read

$$\sin \theta = \frac{\text{opposite}}{1} = \text{opposite}, \quad \cos \theta = \frac{\text{adjacent}}{1} = \text{adjacent},$$

and those two sides are exactly the coordinates of P . So for acute angles the new definition agrees with the old one.

The new form never mentions a triangle. A point can sit anywhere on the circle, at any angle, and it always has coordinates. That is what extends the definition to every real θ .

DEF On the unit circle, for a point P at angle θ :

$$\cos \theta = x\text{-coordinate of } P, \quad \sin \theta = y\text{-coordinate of } P.$$

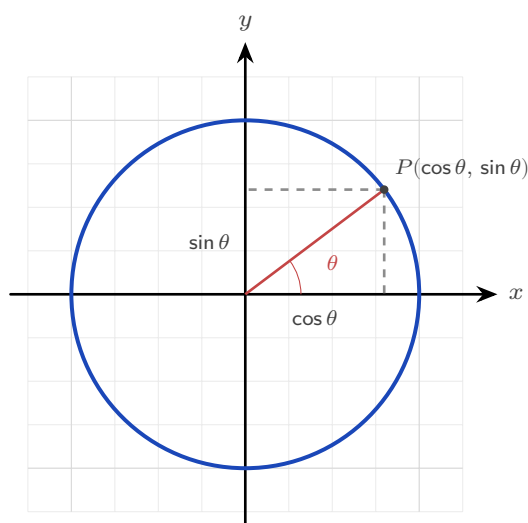


Figure 9.1 On a circle of radius 1 the two shorter sides of the right triangle *are* the coordinates of P , so $\cos \theta$ is read across and $\sin \theta$ up.

The tangent is the gradient of the line OP , so $\tan \theta = \frac{\text{rise}}{\text{run}} = \frac{\sin \theta}{\cos \theta}$. It is undefined wherever $\cos \theta = 0$.

KEY · IB $\tan \theta = \frac{\sin \theta}{\cos \theta}$.

Exact values

A few angles give exact coordinates, and they come from the 30–60–90 and 45–45–90 triangles. These are worth learning by heart.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—

The first-quadrant values repeat into the other three quadrants with signs changed. Collecting them all onto one circle gives the diagram to learn.

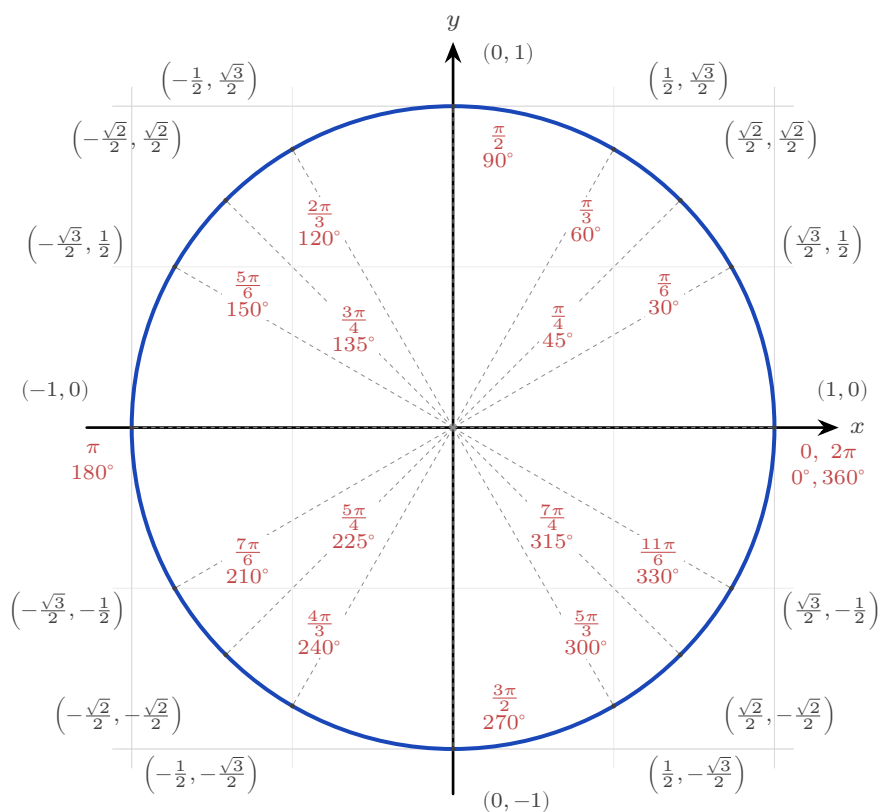


Figure 9.2 The unit circle with every special angle. The scarlet label inside the circle is the angle in radians; the dark label outside is the point $(\cos \theta, \sin \theta)$. To read a value, find the angle and take the matching coordinate.

Notice how little there is to learn. Only three pairs of numbers appear, $(\frac{\sqrt{3}}{2}, \frac{1}{2})$, $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$ and $(\frac{1}{2}, \frac{\sqrt{3}}{2})$, together with the four points on the axes. Everything else is one of those with a sign changed.

Symmetry and the sign of each function

Since $\cos \theta$ and $\sin \theta$ are the coordinates of P , their signs are just the signs of x and y in each quadrant. Nothing needs to be memorised separately: in the second quadrant x is negative and y is positive, so $\cos \theta < 0$ and $\sin \theta > 0$, and $\tan \theta = \frac{\sin}{\cos}$ is negative.

Quadrant	Angles	(x, y)	Positive there	Mnemonic
First	0 to $\frac{\pi}{2}$	$(+, +)$	all three	All
Second	$\frac{\pi}{2}$ to π	$(-, +)$	sin only	Students
Third	π to $\frac{3\pi}{2}$	$(-, -)$	tan only	Take
Fourth	$\frac{3\pi}{2}$ to 2π	$(+, -)$	cos only	Calculus

Reading the last column anticlockwise from the first quadrant gives **All Students Take Calculus**, which is the usual way to remember the pattern. Use it as a check, not as the reason: the reason is always the sign of the coordinates.

Each symmetry of the circle now gives an identity, and each one is a statement about where a reflected point lands.

Reflect in the y -axis. The point at angle $\pi - \theta$ is the mirror image, across the y -axis, of the point at angle θ . Reflecting in the y -axis negates the x -coordinate and leaves y alone, so the two points are $(-\cos \theta, \sin \theta)$ and $(\cos \theta, \sin \theta)$. Comparing coordinates:

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta.$$

Reflect in the x -axis. Measuring $-\theta$ means turning clockwise instead of anticlockwise, which lands on the mirror image of P across the x -axis. That reflection negates y and leaves x alone, so

$$\cos(-\theta) = \cos \theta, \quad \sin(-\theta) = -\sin \theta.$$

Turn a full circle. Adding 2π to the angle sends P once round and back to exactly the

same point, so both coordinates are unchanged:

$$\sin(\theta + 2\pi) = \sin \theta, \quad \cos(\theta + 2\pi) = \cos \theta.$$

This is why both functions are **periodic** with period 2π . Every identity of this kind says the same thing: move the point, see which coordinate changes sign.

You have met the first of these relations before. Two angles, θ and $\pi - \theta$, share the same sine. That shared value is what produced the ambiguous case of the sine rule in Ch. 8. A triangle built from a sine value can sometimes close in two ways, one acute and one obtuse.

YOUR TURN 9.1 The Unit Circle

1. Write down the exact value of each.

- | | | | |
|--------------------------|-----|--------------------------|-----|
| (a) $\sin \frac{\pi}{6}$ | [1] | (b) $\cos \frac{\pi}{4}$ | [1] |
| (c) $\tan \frac{\pi}{3}$ | [1] | (d) $\tan \frac{\pi}{4}$ | [1] |

2. Use the symmetry of the circle to find each exact value.

- | | | | |
|---------------------------|-----|---------------------------|-----|
| (a) $\sin \frac{5\pi}{6}$ | [2] | (b) $\cos \frac{2\pi}{3}$ | [2] |
| (c) $\sin \frac{4\pi}{3}$ | [2] | (d) $\cos \frac{7\pi}{4}$ | [2] |

3. State the quadrant or quadrants in which each condition holds.

- | | | | |
|-----------------------|-----|---|-----|
| (a) $\sin \theta > 0$ | [1] | (b) $\cos \theta < 0$ | [1] |
| (c) $\tan \theta > 0$ | [2] | (d) $\sin \theta < 0$ and $\cos \theta > 0$ | [2] |

4. The point P on the unit circle has angle θ .

- | | |
|--|-----|
| (a) State the coordinates of P in terms of θ . | [1] |
| (b) Explain why $\cos^2 \theta + \sin^2 \theta = 1$ for every θ . | [2] |
| (c) Explain why $\tan \theta$ is undefined at $\theta = \frac{\pi}{2}$. | [2] |

9.2 Trigonometric Identities

An **identity** is an equation true for every angle. The most fundamental one is Pythagoras' theorem read off the unit circle: the point $(\cos \theta, \sin \theta)$ is distance 1 from the origin, so $\cos^2 \theta + \sin^2 \theta = 1$.

KEY · IB $\cos^2 \theta + \sin^2 \theta = 1.$

Double and compound angles

The functions of a sum of two angles can be expanded, and from the sum formulas everything else follows. These **compound-angle** identities are usually just quoted, but the proof is short and uses only the unit circle and the cosine rule from Ch. 8.

Put two points on the unit circle, P at angle A and Q at angle B . Their coordinates are $P(\cos A, \sin A)$ and $Q(\cos B, \sin B)$, and the angle $\angle POQ$ at the centre is $A - B$.

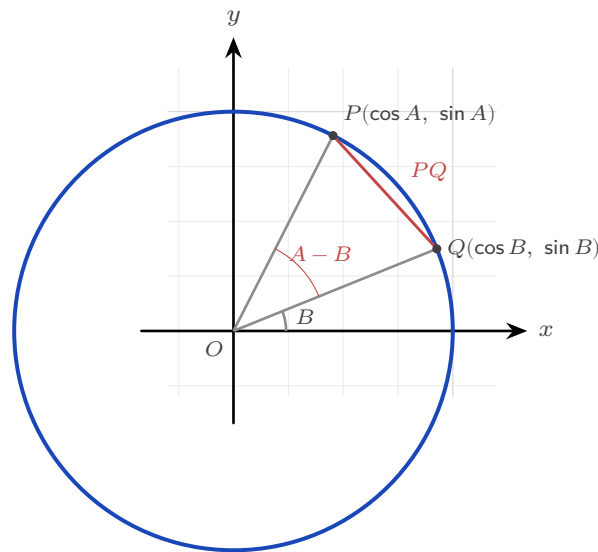


Figure 9.3 Two points on the unit circle, at angles A and B . The angle between the radii is $A - B$, and the chord PQ can be measured either by the distance formula or by the cosine rule.

Now find the distance PQ in two different ways.

By the distance formula:

$$PQ^2 = (\cos A - \cos B)^2 + (\sin A - \sin B)^2.$$

Expanding gives $\cos^2 A - 2 \cos A \cos B + \cos^2 B + \sin^2 A - 2 \sin A \sin B + \sin^2 B$. Both Pythagorean pairs collapse to 1, so

$$PQ^2 = 2 - 2(\cos A \cos B + \sin A \sin B).$$

By the cosine rule in triangle OPQ , where both sides from the centre have length 1 and the angle between them is $A - B$:

$$PQ^2 = 1^2 + 1^2 - 2(1)(1) \cos(A - B) = 2 - 2 \cos(A - B).$$

The two expressions are the same length, so

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Every other compound-angle formula follows from this one. Replacing B by $-B$, and using $\cos(-B) = \cos B$ with $\sin(-B) = -\sin B$, gives the formula for $\cos(A + B)$. The sine versions come from $\sin \theta = \cos(\frac{\pi}{2} - \theta)$, and the tangent versions from dividing the sine formula by the cosine formula.

KEY · IB HL Compound angles:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B, \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

Setting $B = A$ reduces each of these to a **double-angle** identity, the most useful family in the topic. The derivation is worth doing once, because it shows there is nothing new to learn here.

Put $B = A$ in the sine formula:

$$\sin 2A = \sin(A + A) = \sin A \cos A + \cos A \sin A = 2 \sin A \cos A.$$

Do the same in the cosine formula:

$$\cos 2A = \cos(A + A) = \cos A \cos A - \sin A \sin A = \cos^2 A - \sin^2 A.$$

That is the first form. The other two come from $\sin^2 A + \cos^2 A = 1$, used to remove one function or the other. Replacing $\sin^2 A$ by $1 - \cos^2 A$ gives

$$\cos 2A = \cos^2 A - (1 - \cos^2 A) = 2 \cos^2 A - 1,$$

and replacing $\cos^2 A$ by $1 - \sin^2 A$ instead gives

$$\cos 2A = (1 - \sin^2 A) - \sin^2 A = 1 - 2 \sin^2 A.$$

The tangent version follows in the same way, by putting $B = A$ in the tangent formula.

The sine and cosine versions sit in the SL half of the booklet; the tangent version is HL.

KEY · IB Double angles:

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

KEY · IB **HL** $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$

The three forms of $\cos 2\theta$ are one identity rewritten by $\sin^2 + \cos^2 = 1$. Choose whichever leaves you with only sines or only cosines. That choice is what makes the identity so useful when solving equations.

EXAMPLE 9.1

Find the exact value of $\sin 75^\circ$.

Write $75^\circ = 45^\circ + 30^\circ$ and use the compound-angle formula:

$$\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Splitting an awkward angle into a sum of two known ones is the standard method for exact values beyond the basic table.

Identities also let you move between ratios *without ever finding the angle*. Examiners set this often: the angle stays unknown from start to finish, and the Pythagorean identity supplies everything needed.

EXAMPLE 9.2

Given that $\cos x = \frac{3}{4}$ and x is acute, find the exact value of $\sin 2x$ without finding x .

From $\cos^2 x + \sin^2 x = 1$, $\sin^2 x = 1 - \frac{9}{16} = \frac{7}{16}$, and since x is acute, $\sin x = \frac{\sqrt{7}}{4}$ (positive).

Then

$$\sin 2x = 2 \sin x \cos x = 2 \cdot \frac{\sqrt{7}}{4} \cdot \frac{3}{4} = \frac{3\sqrt{7}}{8}.$$

The word “acute” fixed the sign of $\sin x$. If x had instead been in the fourth quadrant, the same working would give $\sin 2x = -\frac{3\sqrt{7}}{8}$: always let the quadrant choose the sign before doubling.

Proving an identity is the same game played in reverse: start from one side, substitute known identities, and simplify until the other side appears.

EXAMPLE 9.3

Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$.

Work on the left side, choosing the form $\cos 2\theta = 2 \cos^2 \theta - 1$ because it cancels the 1:

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2 \sin \theta \cos \theta}{1 + (2 \cos^2 \theta - 1)} = \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta. \quad \blacksquare$$

Choosing the form of $\cos 2\theta$ that cancels the constant is the same idea as in equation solving. Pick the version that matches the terms beside it.

YOUR TURN 9.2 Trigonometric Identities**5.** Simplify each expression.

(a) $\sin^2 \theta + \cos^2 \theta$	[1]	(b) $2 \sin \theta \cos \theta$	[1]
(c) $\cos^2 \theta - \sin^2 \theta$	[1]	(d) $1 - 2 \sin^2 \theta$	[1]

6. Given $\sin \theta = \frac{3}{5}$ with θ acute, find each exact value.

(a) $\cos \theta$	[2]	(b) $\tan \theta$	[1]
(c) $\sin 2\theta$	[2]	(d) $\cos 2\theta$	[2]

7. Given $\cos \theta = -\frac{3}{5}$ with θ in the second quadrant, find each exact value.

(a) $\sin \theta$	[2]	(b) $\tan \theta$	[2]
(c) $\sin 2\theta$	[2]	(d) $\cos 2\theta$	[2]

8. Use a compound-angle formula to find each exact value.

(a) $\cos 15^\circ$, from $45^\circ - 30^\circ$	[3]	(b) $\sin 75^\circ$, from $45^\circ + 30^\circ$	[3]
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9. Prove each identity.

(a) $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$	[3]
(b) $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$	[3]

9.3 Graphs of Trigonometric Functions

The graphs are the circle unrolled. Send the point P round the circle and plot its y -coordinate against the angle: that plot is $y = \sin x$. Plot its x -coordinate instead and you get $y = \cos x$.

Every feature of the graph can be read off the circle before you plot anything.

On the circle	On the graph of $y = \sin x$
the y -coordinate runs from -1 to 1	the curve stays between -1 and 1 : amplitude 1
P returns to its start after one turn	the curve repeats every 2π : period 2π
$y = 0$ at $\theta = 0, \pi, 2\pi$	the curve crosses the axis at $0, \pi, 2\pi$
y is largest at the top, $\theta = \frac{\pi}{2}$	maximum 1 at $x = \frac{\pi}{2}$
$y > 0$ in the upper half of the circle	the curve is above the axis for $0 < x < \pi$

Read the x -coordinate instead and the cosine graph appears the same way. It starts at 1 , because P starts on the positive x -axis, and falls to 0 at $\frac{\pi}{2}$, where P is at the top and its x -coordinate is zero.

That also explains why the two curves have the same shape. The x -coordinate at angle θ is the y -coordinate a quarter turn later, so $\cos \theta = \sin(\theta + \frac{\pi}{2})$, and the cosine curve is the sine curve shifted left by $\frac{\pi}{2}$.

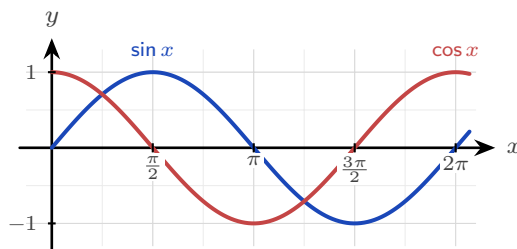


Figure 9.4 One turn of the circle unrolled. Both waves have amplitude 1 and period 2π ; the cosine is the sine shifted left by $\frac{\pi}{2}$.

The tangent behaves differently, and the circle explains why. Since $\tan \theta = \frac{y}{x}$, it is undefined wherever $x = 0$, which is at the top and the bottom of the circle. That gives vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.

Its period is π , not 2π . Half a turn sends P to the point diametrically opposite, whose coordinates are $(-x, -y)$. The ratio $\frac{-y}{-x}$ equals $\frac{y}{x}$, so the tangent repeats after only half a revolution.

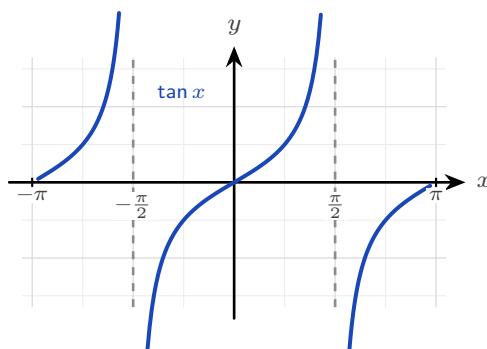


Figure 9.5 The tangent repeats every π , not every 2π , with a vertical asymptote wherever $\cos x = 0$.

Transforming the wave

The general sinusoid stretches and slides the basic sine, and each parameter has a plain meaning.

KEY For $y = a \sin(b(x + c)) + d$:

amplitude = $|a|$, period = $\frac{2\pi}{|b|}$, horizontal shift = $-c$, vertical shift (midline) = d .

Questions often run the other way, giving you the graph's largest and smallest values and asking for a and d . Since the maximum is $d + |a|$ and the minimum is $d - |a|$, adding and subtracting recovers both constants.

Quantity	From the equation	From max and min
Amplitude	$ a $	$\frac{\max - \min}{2}$
Midline	$y = d$	$y = \frac{\max + \min}{2}$
Maximum	$d + a $	—
Minimum	$d - a $	—
Period	$\frac{2\pi}{ b }$	the distance between repeats

The two right-hand formulas are worth reading as sentences. The amplitude is half the total rise and fall, and the midline sits halfway between the top and the bottom. For $y = 4 \sin(3x) - 2$ the maximum is 2 and the minimum is -6 , so the amplitude is $\frac{2 - (-6)}{2} = 4$ and the midline is $\frac{2 + (-6)}{2} = -2$, which matches the equation.

EXAMPLE 9.4

Describe $y = 3 \sin(2x) - 1$ and state its range.

The amplitude is 3, so the wave rises and falls 3 either side of its midline. The period is $\frac{2\pi}{2} = \pi$, so it completes a full cycle twice as fast as $\sin x$. The midline is $y = -1$. The graph therefore runs from $-1 - 3 = -4$ up to $-1 + 3 = 2$, giving range $-4 \leq y \leq 2$.

The four parameters are not just labels on a graph; they are how a sinusoid is fitted to the real world. The opening story can now be settled with actual numbers.

EXAMPLE 9.5

When it opened in 1999, the Daikanransha in Odaiba, Tokyo, was briefly the world's tallest Ferris wheel: 115 m tall, its wheel 100 m across, turning once every 16 minutes. It lost the record to the London Eye within the year, and it was taken down in 2022, but its numbers make a clean model. A seat starts at the bottom of the wheel. Write the seat's height $h(t)$ in metres after t minutes, and find its height after 5 minutes.

Read each parameter off the wheel. The axle sits at $115 - 50 = 65$ m, so the midline is $d = 65$. The radius gives the amplitude, $a = 50$. One revolution takes 16 minutes, so $b = \frac{2\pi}{16} = \frac{\pi}{8}$. The seat starts at the *bottom*, 15 m up, which is what an upside-down cosine does at $t = 0$:

$$h(t) = 65 - 50 \cos\left(\frac{\pi t}{8}\right).$$

Check the skeleton before trusting it: $h(0) = 65 - 50 = 15$ (bottom) and $h(8) = 65 + 50 = 115$ (top, half a turn later). Then

$$h(5) = 65 - 50 \cos \frac{5\pi}{8} \approx 65 + 19.1 \approx 84.1 \text{ m.}$$

Choosing $-\cos$ to start at the minimum, rather than forcing a phase shift into a sine, is the modelling move examiners reward: match the *shape* of the motion first, then the numbers.

Symmetry of the graphs HL

Each reflection symmetry of the circle from Section 9.1 becomes a visible symmetry of the graphs. The cosine curve is mirror-symmetric in the y -axis, so $\cos(-x) = \cos x$: cosine is an **even** function. The sine and tangent curves map onto themselves under a half-turn about the origin, so $\sin(-x) = -\sin x$ and $\tan(-x) = -\tan x$: both are **odd**, in the sense of Ch. 5. The same circle symmetries give the supplementary-angle relations

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta, \quad \tan(\pi - \theta) = -\tan \theta,$$

which you can see on the graphs directly: the sine curve is symmetric about the vertical line $x = \frac{\pi}{2}$, while cosine and tangent flip sign across it. In an exam these relations save you from ever reaching for a calculator: $\cos \frac{3\pi}{4} = -\cos \frac{\pi}{4} = -\frac{\sqrt{2}}{2}$ in one step.

YOUR TURN 9.3 Graphs of Trigonometric Functions

10. State the amplitude, the period and the midline of each graph.

- | | | | |
|--------------------------|-----|--------------------------|-----|
| (a) $y = 3 \sin x$ | [2] | (b) $y = \sin 2x$ | [2] |
| (c) $y = 4 \sin(3x) - 2$ | [3] | (d) $y = 5 \sin(4x) + 3$ | [3] |

11. State the range of each function.

- | | | | |
|--|-----|--------------------------|-----|
| (a) $y = 2 \sin x + 1$ | [2] | (b) $y = 3 \sin(2x) - 1$ | [2] |
| (c) $y = 2 \cos\left(\frac{x}{2}\right) - 1$ | [3] | | |

12. Answer each question about $y = \tan x$.

- | | | | |
|---|-----|---|-----|
| (a) State its period. | [1] | (b) State the equations of two of its asymptotes. | [2] |
| (c) Explain why the asymptotes occur where they do. | [2] | | |

13. A wheel of diameter 40 m has its lowest point 2 m above the ground and turns once every 20 seconds. A seat starts at the bottom.

- | | |
|---|-----|
| (a) State the amplitude and the midline of the seat's height. | [2] |
| (b) Find the value of b in $h(t) = d - a \cos(bt)$. | [2] |
| (c) Write down $h(t)$. | [2] |
| (d) Find the height after 5 seconds. | [2] |

9.4 Reciprocal Trigonometric Functions HL

At Higher Level three more functions are added, the reciprocals of the first three. They introduce no new mathematics: each is one divided by a function you already know. What they do introduce is a set of graphs with asymptotes, and two identities that make certain equations solvable.

KEY · IB $\sec \theta = \frac{1}{\cos \theta}, \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta}.$

The third, the **cotangent**, is $\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$. Its definition is not printed in the booklet, so learn it.

Two facts follow from the reciprocal alone, and between them they fix every graph.

First, a reciprocal has an asymptote wherever the original function is zero, since you cannot divide by zero. So $\sec$ has asymptotes where $\cos \theta = 0$, at $\theta = \frac{\pi}{2} + n\pi$, and cosec and $\cot$ have them where $\sin \theta = 0$, at $\theta = n\pi$.

Second, $\sin$ and $\cos$ never exceed 1 in size, so their reciprocals are never smaller than 1 in size. The range of $\sec \theta$ and of $\operatorname{cosec} \theta$ is therefore $y \leq -1$ or $y \geq 1$: neither ever takes a value strictly between -1 and 1 . This is the fact examiners test most often about these functions.

Function	Definition	Asymptotes at (n any integer)	Range
$\sec \theta$	$\frac{1}{\cos \theta}$	$\theta = \frac{\pi}{2} + n\pi$	$y \leq -1$ or $y \geq 1$
$\operatorname{cosec} \theta$	$\frac{1}{\sin \theta}$	$\theta = n\pi$	$y \leq -1$ or $y \geq 1$
$\cot \theta$	$\frac{\cos \theta}{\sin \theta}$	$\theta = n\pi$	all real y

The two Pythagorean variants

The identity $\cos^2 \theta + \sin^2 \theta = 1$ has two companions, one for each reciprocal pair, and both come from dividing it through.

Dividing $\cos^2 \theta + \sin^2 \theta = 1$ by $\cos^2 \theta$ gives

$$1 + \tan^2 \theta = \sec^2 \theta,$$

since $\frac{\sin \theta}{\cos \theta} = \tan \theta$ and $\frac{1}{\cos \theta} = \sec \theta$. Dividing the same identity by $\sin^2 \theta$ instead gives the companion result. Deriving them takes one line each, which is quicker than trusting your memory of which is which.

KEY · IB $1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta.$

These identities exist to turn mixed equations into single-function ones, exactly as the double angles did.

EXAMPLE 9.6

Solve $\tan^2 x + \sec x = 1$ for $0 \leq x \leq 2\pi$.

The equation mixes $\tan$ and $\sec$; the identity $\tan^2 x = \sec^2 x - 1$ rewrites it in $\sec x$ alone:

$$\sec^2 x - 1 + \sec x = 1 \implies \sec^2 x + \sec x - 2 = 0 \implies (\sec x + 2)(\sec x - 1) = 0.$$

So $\sec x = -2$ or $\sec x = 1$, which is $\cos x = -\frac{1}{2}$ or $\cos x = 1$. From the circle,

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{or} \quad x = 0, 2\pi.$$

Converting back to $\cos$ at the end is the safest final step, because the unit circle is read in sine and cosine, not in secant.

YOUR TURN 9.4 Reciprocal Trigonometric Functions

HL

14. Evaluate each exactly.

- | | | | |
|--------------------------|-----|--|-----|
| (a) $\sec \frac{\pi}{3}$ | [1] | (b) $\operatorname{cosec} \frac{\pi}{2}$ | [1] |
| (c) $\cot \frac{\pi}{4}$ | [1] | (d) $\sec \frac{2\pi}{3}$ | [2] |

15. Simplify each expression.

- | | | | |
|-------------------------------------|-----|---|-----|
| (a) $\sec^2 \theta - \tan^2 \theta$ | [1] | (b) $\operatorname{cosec}^2 \theta - \cot^2 \theta$ | [2] |
| (c) $\frac{1}{\cot \theta}$ | [1] | (d) $\tan \theta \cot \theta$ | [2] |

9.5 Inverse Trigonometric Functions

HL

Everything so far has gone one way: give an angle, get a ratio. The inverse functions run the other way, recovering the angle from the ratio.

You have already used them without naming them. When you solve $\sin x = \frac{1}{2}$ and press the $\sin^{-1}$ key, that key is the inverse function.

Two notations, one function. The inverse of sine is written either $\arcsin x$ or $\sin^{-1} x$.

They mean exactly the same thing. This book uses $\arcsin$, because the other form invites a serious confusion:

$$\sin^{-1} x \text{ means the inverse,} \quad \text{but} \quad (\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x.$$

The -1 is not an exponent here. Writing $\arcsin$, $\arccos$ and $\arctan$ removes the ambiguity, and your calculator's $\sin^{-1}$ key is the same function under the other name.

Why a restriction is needed

There is an immediate difficulty. A function has an inverse only if it is one-to-one (Ch. 5), and $\sin$, $\cos$ and $\tan$ are as far from one-to-one as a function can be. Each takes every value in its range infinitely often, so the horizontal line $y = \frac{1}{2}$ meets $y = \sin x$ infinitely many times.

That is a real problem, not a technicality. If you ask “which angle has sine $\frac{1}{2}$?”, there is no single answer: $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{13\pi}{6}$ and infinitely many others all qualify. A function must return one output, so it has to choose.

The fix is the one from Ch. 5: cut the domain down until the function is one-to-one, then invert what is left. Which piece to keep is a choice, and mathematicians have agreed on one for each function. Those agreed pieces are shaded in the diagrams below, and they are what fix the ranges in the table.

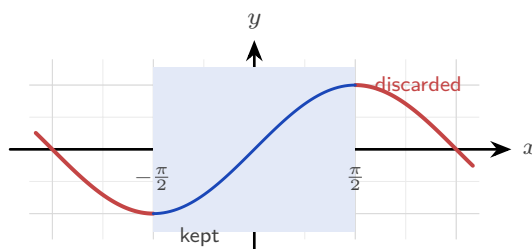


Figure 9.6 To invert $y = \sin x$, keep only the shaded piece from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. On that piece the curve rises steadily, so it is one-to-one and each output comes from exactly one input.

The same is done for the other two: cosine keeps $0 \leq x \leq \pi$, where it falls steadily, and tangent keeps $-\frac{\pi}{2} < x < \frac{\pi}{2}$, one branch between two asymptotes. Those kept pieces become the *ranges* of the inverses, because domain and range swap when a function is inverted.

Function	Domain	Range
$\arcsin x$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
$\arccos x$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
$\arctan x$	all real x	$-\frac{\pi}{2} < y < \frac{\pi}{2}$

Each inverse is a reflection of its restricted parent in the line $y = x$ (Ch. 5), so the graphs are the familiar curves turned on their sides. Arcsine climbs steeply from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$. Arccosine falls from $(-1, \pi)$ to $(1, 0)$. Arctangent flattens towards the horizontal asymptotes $y = \pm\frac{\pi}{2}$, which are the vertical asymptotes of tangent after the reflection.

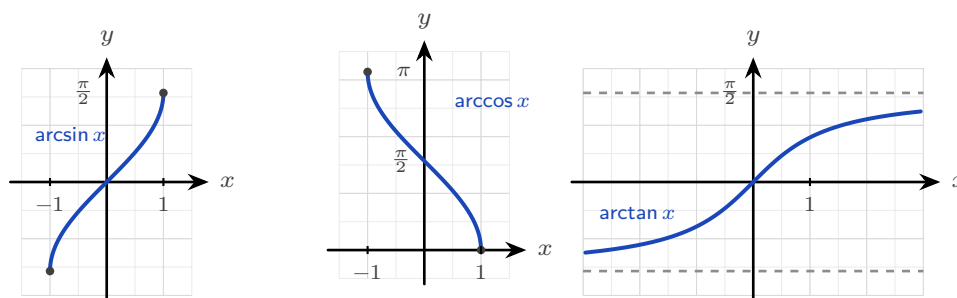


Figure 9.7 Each inverse is its restricted parent reflected in $y = x$. The marked endpoints show the range each one is confined to.

EXAMPLE 9.7

Find the exact values of (a) $\arcsin(-\frac{1}{2})$, (b) $\arccos(-\frac{\sqrt{3}}{2})$, (c) $\arcsin(\sin \frac{5\pi}{6})$.

(a) The angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is $-\frac{1}{2}$ is $-\frac{\pi}{6}$. Not $\frac{11\pi}{6}$ and not $\frac{7\pi}{6}$: those solve the equation but lie outside arcsine's range.

(b) The angle in $[0, \pi]$ whose cosine is $-\frac{\sqrt{3}}{2}$ is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Arccosine handles negatives by reaching into the second quadrant.

(c) $\sin \frac{5\pi}{6} = \frac{1}{2}$, and $\arcsin \frac{1}{2} = \frac{\pi}{6}$. The composition returns $\frac{\pi}{6}$, not $\frac{5\pi}{6}$: arcsine can only answer with angles from its own range.

CAUTION $\arcsin(\sin x)$ does not always return x . Because arcsin only outputs angles in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, feeding it $\sin(2)$ gives $\pi - 2$, not 2. The restricted range is doing exactly its job.

YOUR TURN 9.5 Inverse Trigonometric Functions

HL

16. Evaluate each exactly.

(a) $\arcsin \frac{1}{2}$ [1] (b) $\arccos 0$ [1]

(c) $\arctan(-1)$ [2] (d) $\arccos(-\frac{1}{2})$ [2]

17. These compositions do not simply cancel.

(a) State the range of $\arcsin x$ and the range of $\arccos x$. [2]

(b) Evaluate $\arcsin(\sin \frac{3\pi}{4})$, and explain why the answer is not $\frac{3\pi}{4}$. [3]

(c) Evaluate $\arccos(\cos \frac{4\pi}{3})$. [3]

9.6 Solving Trigonometric Equations

An equation like $\sin x = \frac{1}{2}$ has not one solution but infinitely many, because the wave crosses any horizontal line over and over. Your calculator returns only one of them, the **principal value**. Everything else you must find yourself, and the circle is what tells you how.

Here is the reason there is always a second angle. Solving $\sin x = k$ means finding the points on the circle whose y -coordinate is k . A horizontal line at height k cuts the circle in *two* places, not one, so two angles share that sine. The same is true for $\cos x = k$, where a vertical line at $x = k$ cuts the circle twice.

Where the second angle sits depends on which function you have. The rule is short, and it is worth learning as one line.

Equation	Second angle
$\sin x = k$	$\pi - \alpha$
$\cos x = k$	$2\pi - \alpha$
$\tan x = k$	$\alpha + \pi$

Each line is a symmetry of the circle. For sine the two points sit at the same height, mirrored in the y -axis. For cosine they sit at the same horizontal position, mirrored in the x -axis. For tangent the second point is diametrically opposite, half a turn away.

Then check the signs. Your two angles should lie in the two quadrants where that function is positive, or in the other two if k is negative. If only one of them does, you have used the wrong row.

Once you have both angles in one turn, add or subtract whole turns of 2π to reach every solution the interval asks for.

KEY Solving a trigonometric equation.

1. Rearrange until a single trigonometric function is alone on one side.
2. If the argument is not plain x , substitute u for it and **transform the interval as well**.
3. Find the principal value α from your calculator.
4. Find the second angle from the table above.
5. Add or subtract 2π (or $2\pi b$, in terms of u) until you have every solution in the interval.
6. If you substituted, convert back to x at the very end.

Step 2 is where marks are most often lost, and step 5 is where solutions go missing.

EXAMPLE 9.8

Solve $\cos x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

The calculator gives the principal value $\alpha = \arccos(-\frac{1}{2}) = \frac{2\pi}{3}$.

This is a cosine equation, so the partner angle is $2\pi - \alpha$:

$$2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}.$$

Both lie in the interval, so $x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.

Check the signs against the circle. Cosine is negative in the second and third quadrants, and $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$ are exactly one angle from each. That check catches a wrong partner immediately.

EXAMPLE 9.9

Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.

The principal value is $x = \frac{\pi}{6}$. Sine is also positive in the second quadrant, where the partner angle is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Both lie in the interval, so

$$x = \frac{\pi}{6} \quad \text{or} \quad x = \frac{5\pi}{6}.$$

Always ask which *other* quadrant shares the sign; that second angle is the one candidates most often miss.

TIP On a calculator paper, always check your solution count graphically: plot both sides and count the intersections in the interval. If your algebra found three solutions and the screen shows four, the screen is right.

When the argument is not plain x but $2x$ or $3(x-4)$, substitute for the whole argument and, crucially, transform the interval along with it. A doubled argument sweeps the circle twice as fast, so it collects twice as many solutions.

EXAMPLE 9.10

Solve $\sin 2x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.

Let $u = 2x$. As x runs over $[0, 2\pi]$, u runs over $[0, 4\pi]$: two full turns. In the first turn, $\sin u = \frac{1}{2}$ at $u = \frac{\pi}{6}$ and $\frac{5\pi}{6}$; the second turn repeats them 2π later, at $\frac{13\pi}{6}$ and $\frac{17\pi}{6}$. Halving each u recovers x :

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}.$$

Four solutions, not two. Solving in u over the *enlarged* interval first, and only then dividing, is what protects you from losing the second pair.

Using identities

An equation mixing $\sin 2x$ with $\sin x$, or $\cos 2x$ with $\cos x$, cannot be solved as it stands. An identity rewrites everything in terms of a single function, turning the equation into an algebraic one, usually a quadratic.

EXAMPLE 9.11

Solve $\cos 2x + \cos x = 0$ for $0 \leq x \leq 2\pi$.

Choose the form of $\cos 2x$ that gives only cosines, $\cos 2x = 2 \cos^2 x - 1$:

$$2 \cos^2 x - 1 + \cos x = 0 \implies 2 \cos^2 x + \cos x - 1 = 0.$$

This factors as $(2 \cos x - 1)(\cos x + 1) = 0$, so $\cos x = \frac{1}{2}$ or $\cos x = -1$. From the circle,

$$x = \frac{\pi}{3}, \frac{5\pi}{3} \quad \text{or} \quad x = \pi.$$

Picking the form of the double-angle identity that matches the other terms is the key step in these problems.

Enrichment: the wave form $a \sin \theta + b \cos \theta$

This last idea sits just beyond the AA syllabus. It is included because it is both elegant and useful. A sum of a sine and a cosine of the same angle is itself a single shifted sine wave: the compound-angle identity, run backwards. Writing it as $R \sin(\theta + \alpha)$ makes both its maximum and its equation solvable at a glance.

KEY $a \sin \theta + b \cos \theta = R \sin(\theta + \alpha)$, where $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$. The maximum value is R , the minimum $-R$.

A related idea appears when a problem asks for *all* solutions rather than those in a fixed interval. The answer is then a family generated by the period. For example, $\sin x = k$ has general solution $x = \arcsin k + 2n\pi$ or $x = (\pi - \arcsin k) + 2n\pi$, for every integer n . The AA syllabus never asks for general solutions in an exam, but seeing the infinite family explains why a question must always restrict you to an interval.

YOUR TURN 9.6 Solving Trigonometric Equations

18. Solve for $0 \leq x \leq 2\pi$.

(a) $\sin x = \frac{1}{2}$

[2]

(b) $\cos x = 0$

[2]

(c) $\tan x = 1$

[2]

(d) $\cos x = 1$

[1]

19. Solve for $0 \leq x \leq 2\pi$.

- | | | | |
|-----------------------------|-----|---------------------------|-----|
| (a) $\sin x = -\frac{1}{2}$ | [2] | (b) $2 \cos x = \sqrt{3}$ | [2] |
| (c) $\sqrt{2} \sin x = -1$ | [3] | (d) $\tan x = -\sqrt{3}$ | [3] |

20. Solve for $0 \leq x \leq 2\pi$. Transform the interval before solving.

- | | | | |
|-------------------|-----|-----------------------------|-----|
| (a) $\sin 2x = 0$ | [2] | (b) $\cos 2x = \frac{1}{2}$ | [3] |
| (c) $\tan 2x = 1$ | [4] | | |

21. Solve for $0 \leq x \leq 2\pi$, using an identity first.

- | | |
|-----------------------------------|-----|
| (a) $2 \sin^2 x - \sin x = 0$ | [3] |
| (b) $2 \cos^2 x + \cos x - 1 = 0$ | [4] |
| (c) $\cos 2x + \sin x = 0$ | [4] |

Reflections

IM Trigonometry began as astronomy. Babylonian and Greek observers needed to predict the wandering of the planets, and the sine was born as a tool for the geometry of the celestial sphere. Its greatest early development came in India and in the Islamic world. Astronomers there tabulated sines to many decimal places, to fix the times of prayer and the direction of Mecca. The subject returned to Europe through Latin translations, which gave us the word “sine” itself. For two thousand years, the functions of this chapter were the mathematics of the heavens.

TOK A sine wave is periodic: it repeats forever, identical in every cycle. Yet it is built from the endlessly irrational π and takes irrational values at almost every point. How can something so regular be made of something so unruly? The physicist Joseph Fourier then proved that *any* repeating signal, however jagged, is a sum of pure sine waves. So the smooth circle is hidden inside every square wave and every spoken word. Is the sinusoid a human tool, or the alphabet in which nature writes all its rhythms?

RW Because a sine wave is what steady rotation looks like from the side, it models every oscillation in science and engineering. Alternating current rises and falls as a sine while the generator spins. Sound is pressure oscillating in the same way, and its pitch is the frequency of the wave. Light, radio, and every signal your phone sends and receives are sinusoids too. Fourier analysis, decomposing a signal into its sine components, is how music is compressed, images are stored, and earthquakes and heartbeats are read.

End-of-Chapter Problems — Chapter 9: Trigonometric Functions

1. Use the unit circle to write down the exact value of each.

- | | | | |
|----------------------------|-----|---------------------------|-----|
| (a) $\sin \frac{2\pi}{3}$ | [1] | (b) $\cos \frac{2\pi}{3}$ | [1] |
| (c) $\tan \frac{2\pi}{3}$ | [2] | (d) $\cos \frac{5\pi}{4}$ | [2] |
| (e) $\sin \frac{11\pi}{6}$ | [2] | (f) $\tan \frac{3\pi}{4}$ | [2] |

2. The point P lies on the unit circle at angle θ .

- (a) State the coordinates of P when $\theta = \frac{5\pi}{6}$. [2]
- (b) P has coordinates $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$. Find θ for $0 \leq \theta \leq 2\pi$. [2]
- (c) P lies in the third quadrant. State the sign of $\sin \theta$, of $\cos \theta$ and of $\tan \theta$. [3]
- (d) Given $\sin \theta = \frac{\sqrt{3}}{2}$, write down the two possible values of θ in $0 \leq \theta \leq 2\pi$, and explain why there are exactly two. [3]

3. Prove each identity. They are arranged in increasing order of difficulty.

- (a) $\cos^2 \theta - \sin^2 \theta = 1 - 2\sin^2 \theta$ [2]
- (b) $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$ [3]
- (c) $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$ [3]
- (d) $\cos^4 \theta - \sin^4 \theta = \cos 2\theta$ [3]
- (e) $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$ [4]
- (f) $\frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta$ [4]
- (g) $\frac{1 + \cos \theta}{1 + \sin \theta} + \frac{1 + \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$ [5]
- (h) $\frac{1 + \sin 2\theta}{\cos 2\theta} = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$ [5]

4. Given $\cos \theta = -\frac{3}{5}$ with θ in the second quadrant, find the exact value of each.

- | | | | |
|--------------------|-----|--------------------|-----|
| (a) $\sin \theta$ | [2] | (b) $\tan \theta$ | [1] |
| (c) $\sin 2\theta$ | [2] | (d) $\cos 2\theta$ | [2] |

5. Given $\tan \theta = 2$ with θ acute, find the exact value of each.

- (a) $\sin \theta$ and $\cos \theta$ [2] (b) $\sin 2\theta$ [2]
 (c) $\cos 2\theta$ [2]

6. Use a compound-angle formula to find each exact value.

- (a) $\tan 75^\circ$ [4] (b) $\cos 15^\circ$ [3]

7. Prove each result.

- (a) $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ [4]
 (b) $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ [5]
 (c) $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ [5]

8. Solve for $0 \leq x \leq 2\pi$.

- (a) $\sin x = \frac{1}{2}$ [2] (b) $\sqrt{3} \tan x = 1$ [3]
 (c) $2 \cos^2 x = 1$ [4] (d) $\tan^2 x = 3$ [4]

9. Solve for $0 \leq x \leq 2\pi$. Each becomes a quadratic.

- (a) $2 \sin^2 x + \sin x - 1 = 0$ [5] (b) $4 \sin^2 x - 3 = 0$ [4]
 (c) $\cos 2x = \cos x$ [5] (d) $\sin 2x = \sin x$ [5]

10. Solve for $0 \leq x \leq 2\pi$.

- (a) $2 \sin x \cos x = \frac{1}{2}$ [4]
 (b) $\cos 2x + 3 \sin x = 2$ [6]
 (c) $\sin x + \cos x = 1$ [6]

11. Given the identity $1 + \tan^2 \theta = \sec^2 \theta$, solve $\sec^2 x = 4$ for $0 \leq x \leq \frac{\pi}{2}$. [4]

12. Evaluate, giving exact values.

- (a) $\arctan 1$ [1] (b) $\arcsin\left(-\frac{1}{2}\right)$ [1]
 (c) $\arccos(-1)$ [1] (d) $\arcsin\left(\sin \frac{5\pi}{6}\right)$ [3]

13. Consider $f(x) = 3 \sin(2x) + 1$.

- (a) State the amplitude and the period. [2]
 (b) State the maximum and minimum values. [2]

14. The graph of $y = a \sin(bx) + d$ has maximum 7, minimum 1, and period π . Find a , b

and d . [4]

15. Sketch $y = \cos x - 1$ for $0 \leq x \leq 2\pi$, stating the maximum, the minimum, and the intercepts. [4]

16. The depth of water in a harbour is $d(t) = 5 + 3 \sin\left(\frac{\pi t}{6}\right)$ metres, t hours after midnight.

- (a) State the maximum and minimum depths. [2]
- (b) Find the period, and say what it represents. [2]

17. A seat on a Ferris wheel has height $h(t) = 10 - 8 \cos\left(\frac{\pi t}{15}\right)$ metres, t seconds after the start.

- (a) State the minimum and maximum heights. [2]
- (b) Find the time for one full revolution. [2]

18. Express $3 \sin x + 4 \cos x$ in the form $R \sin(x + \alpha)$ with $R > 0$.

- (a) Find R and α . [3]
- (b) Hence state the maximum value. [1]

19. Find the maximum value M of $5 \sin x + 12 \cos x$, and the value of x in $[0, 2\pi]$ at which it occurs. [5]

20. Use the symmetry of the unit circle to answer each part.

- (a) Given $\sin 0.6 = 0.565$, write down $\sin(\pi - 0.6)$. [2]
- (b) Given $\cos 1.2 = 0.362$, write down $\cos(-1.2)$ and $\cos(2\pi - 1.2)$. [3]
- (c) Explain why $\sin(\theta + 2\pi) = \sin \theta$ for every θ . [2]
- (d) Explain why $\tan(\theta + \pi) = \tan \theta$, but $\sin(\theta + \pi) \neq \sin \theta$. [4]

Challenge Questions

21. The wave form, and when an equation can be solved. Section 9.5 states that $a \sin \theta + b \cos \theta$ can be written as $R \sin(\theta + \alpha)$. This investigation proves it and then uses it to decide, without solving, whether an equation has any solutions at all.

Throughout take $a > 0$ and $b > 0$.

- (a) Expand $R \sin(\theta + \alpha)$ and compare it with $a \sin \theta + b \cos \theta$ to show that $R \cos \alpha = a$ and $R \sin \alpha = b$. [3]
- (b) Hence show that $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$. (*Hint: square and add*) [3]
- (c) Write $7 \sin \theta + 24 \cos \theta$ in the form $R \sin(\theta + \alpha)$, and state its maximum and minimum values. [3]
- (d) Explain why the equation $a \sin \theta + b \cos \theta = k$ has a solution if and only if $|k| \leq R$. [4]
- (e) Solve $3 \sin x + 4 \cos x = 5$ for $0 \leq x \leq 2\pi$, giving your answer to three significant figures. Explain why this equation has exactly one solution in the interval. [5]

22. $\cos n\theta$ as a polynomial in $\cos \theta$. You have already met $\cos 2\theta = 2 \cos^2 \theta - 1$ and, in the end-of-chapter problems, $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. Each is a polynomial in $\cos \theta$ alone. This investigation shows that this always happens, and finds the pattern.

Write $c = \cos \theta$ throughout.

- (a) Use the compound-angle formulas to show that

$$\cos((n+1)\theta) = 2 \cos \theta \cos(n\theta) - \cos((n-1)\theta).$$

(*Hint: expand $\cos(n\theta + \theta)$ and $\cos(n\theta - \theta)$, then add*) [5]

- (b) Starting from $\cos 0 = 1$ and $\cos \theta = c$, use the recurrence to obtain $\cos 2\theta$ and $\cos 3\theta$ in terms of c . [3]
- (c) Use it again to find $\cos 4\theta$ and $\cos 5\theta$ in terms of c . [4]
- (d) State the degree of the polynomial for $\cos n\theta$, and the coefficient of its highest power. Justify both from the recurrence. [4]
- (e) Explain why no such polynomial in $\cos \theta$ alone exists for $\sin n\theta$ when n is even. (*Hint: compare $\sin(-\theta)$ with $\sin \theta$*) [3]

23. Counting solutions before finding them. A common examination error is to lose solutions when the argument is not plain x . This investigation finds how many solutions there are, so that you can check your own answer.

Take b to be a positive integer and consider $\sin(bx) = k$ on the interval $0 \leq x \leq 2\pi$.

- (a) Solve $\sin x = \frac{1}{2}$, then $\sin 2x = \frac{1}{2}$, then $\sin 3x = \frac{1}{2}$, on $0 \leq x \leq 2\pi$. State how many solutions each has. [4]
- (b) Explain why substituting $u = bx$ turns the interval $0 \leq x \leq 2\pi$ into $0 \leq u \leq 2\pi b$. [2]
- (c) Hence state, with reasons, the number of solutions of $\sin(bx) = k$ when $|k| < 1$. [3]
- (d) State the number of solutions when $k = 1$, and when $|k| > 1$. Explain the difference. [4]
- (e) How many solutions does $\cos(bx) = k$ have when $|k| < 1$? Justify your answer, and say whether the endpoints $x = 0$ and $x = 2\pi$ need special care. [4]

